

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Robotics & AI Engineer · Embedded Systems Developer · Edge AI & SciML Builder

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EXECUTIVE PROFILE & RESEARCH DIRECTIVE

High-velocity Robotics and Artificial Intelligence Engineering undergraduate (CGPA: 9.05 / 10.00, Currently in 3rd Year) with proven experience spanning low-level embedded hardware, deterministic edge autonomy, scientific machine learning, and full-stack systems engineering. Lead Author of accepted robotics perception research (IEEE ICRM 2025), Co-Investigator on coastal thermal desalination instrumentation (Elsevier Desalination), and Global Top-5 Finalist in the IEEE IES Generation-AI Challenge 2024. Proven full-stack builder across bare-metal C/C++, Arduino/STM32/ESP32 firmware, ROS 2, AMD Ryzen AI XDNA NPUs, and physics-constrained deep learning models.

ACADEMIC PEDIGREE & UNIVERSITY EDUCATION

Amrita Vishwa Vidyapeetham, School of Engineering

Amritapuri, Kerala, India

Bachelor of Technology (B.Tech) in Robotics and Artificial Intelligence Engineering CGPA: 9.05 / 10.00 · Aug 2024 – May 2028

- Academic Standing: Top 1% percentile of department; currently in 3rd Year of undergraduate studies.
- Academic Honors: Recipient of prestigious University Top Tier Merit Scholarship across consecutive academic years.
- Verified Coursework Breakdown:
 - Pure & Applied Mathematics: Multivariable Calculus (Vector Fields, Green's/Stokes'/Divergence Theorems, Jacobians), Linear Algebra (Vector Spaces, Eigenvalues, SVD, QR Factorization), Ordinary & Partial Differential Equations (Heat/Wave PDEs, Fourier/Laplace Transforms), Complex Analysis, Numerical Methods & Optimization, Probability & Statistics.
 - Robotics, Kinematics & Controls: Spatial Coordinate Transformations (SO(3), SE(3), Quaternions), Forward & Inverse Kinematics, Robot Dynamics (Euler-Lagrange Formulations), Classical PID, State-Space Control (LQR, Pole Placement), State Estimation (EKF/UKF), LiDAR SLAM, Nav2 Stack.
 - Computer Science & Real-Time Systems: Data Structures & Algorithms (Graph Traversal, Dynamic Programming), ARM Cortex-M & Microcontroller Architecture (NVIC, SysTick, DMA), Real-Time Operating Systems (FreeRTOS, Preemptive Scheduling, Mutex/Semaphore IPC), Linux Systems Programming, Digital Signal Processing (DSP, FIR/IIR Filters, FFT), Computer Vision (OpenCV, Spatial Filtering, Hough Transforms).
 - AI & Scientific Machine Learning: Deep Learning Architectures (CNNs, Vision Transformers), Physics-Informed Neural Networks (PINNs, Residual PDE Collocation Loss), Reduced-Order Modeling (POD-Galerkin, SVD), Sparse Identification of Nonlinear Dynamics (SINDy), Reinforcement Learning.

NATIONAL COMPETITIVE HONORS & SCIENTIFIC AWARDS

- Global Finalist (Top-5 Worldwide), IEEE IES Generation-AI Challenge (2024): Selected among 400+ international research teams for pioneering thermodynamic physics-guided generative diffusion models constrained by Gibbs free energy surfaces.
- ISRO YUVIKA Space Science Fellow (2020): Selected in the Top 0.1% nationwide tier by the Indian Space Research Organisation (ISRO) for intensive residential training in satellite payloads, atmospheric telemetry, and rocket propulsion systems.
- National Anveshika Experimental Physics Competition (NAEST 2021): Qualified two consecutive national screening tiers conducted by Padma Shri Prof. H. C. Verma (IIT Kanpur), demonstrating advanced experimental mechanics and optics problem-solving.
- State Rank 497 / 150,000+, AP POLYCET (2022): Achieved Top 0.3% statewide engineering merit rank among over 150,000 candidates in Andhra Pradesh.
- State Finalist, MTSO Mathematics Olympiad (2021): Awarded State Merit Honors for advanced analytical reasoning, combinatorial mathematics, and number theory.

PEER-REVIEWED RESEARCH PUBLICATIONS & PREPRINTS

- Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity
IEEE International Conference on Robotics and Mechatronics (ICRM 2025) Accepted & Presented · Jan 2025
Authors: Nallabati Venkata Durga Sai Sarat Chandra, et al.
 - Engineered real-time edge vision pipeline executing Probabilistic Hough Line Transforms at 35+ FPS on ARM Cortex-A SBCs.
 - Implemented closed-loop PID lateral trajectory steering coupled with zero-copy asynchronous UDP telemetry streaming (<15 ms latency).

2. 25-Day Coastal Field Performance of Solar Parabolic Dish Desalination System Under Tropical Marine Climates
Elsevier Desalination Journal Submission First Author / Co-Investigator · Under Review, 2024
Authors: Nallabati Venkata Durga Sai Sarat Chandra, et al.
 - Conducted exhaustive 25-day field campaign logging solar flux, focal receiver thermal distribution (116°C), and distillate yields.
 - Achieved 99.5% Total Dissolved Solids (TDS) reduction (3,850 to 18mg/L) and modeled empirical correlation (Adjusted $R^2 = 0.83$, $p < 0.05$).
3. Physics-Informed Neural Networks for Non-Newtonian Blood Rheology Inversion Under Pulsatile Hemodynamics
Preprint in Preparation for International Journal for Numerical Methods in Biomedical Engineering 2025
Authors: Nallabati Venkata Durga Sai Sarat Chandra.
 - Formulated Navier-Stokes residual loss network inverting Carreau-Yasuda parameters with <2.4% relative L_2 error vs ANSYS Fluent CFD.

GRANULAR TECHNICAL SKILLS TAXONOMY

- Programming Languages: Modern C++ (C++17/20), Embedded C (C99/C11), Python (Expert: NumPy, SciPy, Pandas, PyTorch), SQL (PostgreSQL, SQLite), TypeScript, JavaScript, MATLAB, R, Rust (Basics), Linux Shell/Bash, LaTeX.
- Robotics & Autonomy Software: ROS 2 (Jazzy, Humble), CycloneDDS Middleware (Zero-Copy Shared Memory), OpenCV, Probabilistic Hough Transforms, 5-DOF Damped Least-Squares (DLS) Inverse Kinematics, Velocity Profiling (S-Curve, Linear PWM Ramping), Nav2 Stack, Gazebo/RViz2.
- Embedded Systems & Microcontrollers: STM32 (ARM Cortex-M), Arduino Uno/Nano, ESP32 Dual-Core Xtensa, ESP8266, FreeRTOS (Preemptive Scheduling, Queues, Mutexes), Bare-Metal Driver Development, ST-Link / J-Link GDB Debugging.
- Hardware Protocols & Low-Level Buses: CAN Bus (2.0B / CAN-FD), Circular DMA UART (921,600 baud), Hardware SPI, I2C with 9-Clock Bus Lockup Auto-Recovery, ADC/DAC DMA Streams, Advanced PWM Timers, Binary File Serialization.
- Scientific ML & Computational AI: Physics-Informed Neural Networks (PINNs), Proper Orthogonal Decomposition (POD / SVD), Sparse Identification of Nonlinear Dynamics (SINDy), ANSYS Fluent Transient CFD, Carreau-Yasuda Rheology, PyTorch, SciPy.
- Edge AI & Silicon Acceleration: AMD Ryzen AI XDNA NPU, ONNX Runtime (Vitis AI Execution Provider), INT8 Post-Training Quantization (PTQ), ROCm 6.x / HIP Heterogeneous Compute Bridges, Raspberry Pi 5 Edge Computing.
- Industrial IoT, Cellular & Hardware: Valetron/SIMCOM A7670C 4G LTE Modem, GSM Non-Blocking AT Command State Machines, Dual-Mode WiFi ↔ 4G Failover, Opto-Isolated 240V Relays, RC Snubber Transient Suppression, KiCAD PCB Layout.
- Full-Stack, Data Science & FinTech: React Native, Node.js, Parametric Insurance Architecture, OpenWeather API, GPS Anomaly Detection, Financial Time-Series Analytics, Optuna Bayesian Optimization, HTML5/Tailwind CSS.

PROFESSIONAL EXPERIENCE & SYSTEMS LEADERSHIP

CLUB MIRAI (Robotics & Embedded Systems Club)

Amrita Vishwa Vidyapeetham

Embedded Systems and Robotics Developer & Hardware Systems Lead

Aug 2023 – Present

- Architected multi-node distributed robotics compute bus integrating Raspberry Pi 5 master with dual STM32/ESP32 slave nodes over CAN 2.0B at 500 kbps and DMA-driven UART at 921,600 baud with zero packet loss under sustained 100Hz telemetry.
- Designed and laid out 4-layer custom motor controller PCBs in KiCAD featuring dedicated power/ground planes, opto-isolated gate drivers, and hardware snubber circuits handling inductive back-EMF spikes up to 480V.
- Mentored and supervised a cohort of 12 junior engineers across schematic capture, FreeRTOS task synchronization, MISRA C defensive coding practices, and CycloneDDS zero-copy middleware configuration.

FACTUAL ENGINEERING & SCIENTIFIC RESEARCH REPOSITORY (VERIFIED ON GITHUB)

Edge AI Systems, Silicon Acceleration & Machine Learning

1. Evolution Edge: Self-Evolving Edge AI Routing Engine

GitHub/Evolution-Edge · 2024

AMD Developer Hackathon · AMD Ryzen AI XDNA NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, PEFT/LoRA

- Engineered dynamic neuro-symbolic routing engine elevating local edge LLMs (Qwen/Llama) via real-time query complexity gating.
- Out-of-distribution queries automatically route to AMD Instinct MI300X cloud teachers, generating Knowledge Packets that update edge semantic memory.
- Deployed quantized INT8 neural models to AMD Ryzen AI NPU via ONNX Runtime Vitis AI EP, achieving <28.4 ms latency within 8W TDP envelope.

2. GigGuard AI: Parametric Weather & Income Insurance Platform [GitHub/GigGuard-AI](#) · 2024
DEVTrails Hackathon · Founder & Lead Architect · React Native, Python, OpenWeather API, Dynamic ML Risk Pricing
- Designed parametric income insurance platform for Q-commerce delivery partners (Zepto/Blinkit/Swiggy) suffering income loss during severe weather.
 - Automated instant payout triggers based on real-time weather APIs (Rainfall > 40 mm/hr, AQI > 350, local municipal curfews).
 - Built dynamic ML risk-based premium pricing model (INR 25/week) and intelligent anomaly-detection engine preventing GPS spoofing fraud.
3. Thermodynamic Physics-Guided Generative Diffusion Models [IEEE Global Finalist](#) · 2024
IEEE IES GenAI Challenge · PyTorch, Equivariant GNNs, Gibbs Free Energy Loss, Score-Based SDEs, SE(3) Invariance
- Developed score-based generative diffusion model embedding thermodynamic equilibrium constraints into reverse-time stochastic differential equations for molecular crystal generation.
 - Achieved Top-5 Finalist Selection out of 400+ international teams in the IEEE Industrial Electronics Society (IES) GenAI Challenge 2024.
 - Benchmarked crystal lattice validity against DFT simulations, demonstrating 96.8% structural physical validity.
4. Physics-Informed Neural Network (PINN) Non-Newtonian Hemodynamics Pipeline Lead Developer · 2024 – Present
PyTorch, SciPy, ANSYS Fluent CFD, Carreau-Yasuda Rheology, Adaptive NTK Trace Balancing, Optuna
- Formulated Navier-Stokes forward/inverse physics-loss PINN inverting blood rheology parameters from pulsatile CFD velocity fields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot [2\mu(\dot{\gamma})\mathbf{D}], \quad \mu(\dot{\gamma}) = \mu_{\infty} + (\mu_0 - \mu_{\infty}) [1 + (\lambda\dot{\gamma})^a]^{(n-1)/a}$$

- Implemented Neural Tangent Kernel (NTK) trace balancing to eliminate gradient pathologies, achieving <2.4% relative L_2 error against high-resolution transient ANSYS Fluent ground truth.
 - Engineered automated PyTorch data ingestion pipeline accelerating hyperparameter sweeps by $4.2\times$ via Optuna Bayesian optimization.
5. Reduced-Order Modeling (ROM) & Sparse Nonlinear Dynamics (SINDy) [Lead Developer](#) · 2024
Proper Orthogonal Decomposition (POD/SVD), STLSQ Sparse Regression, Python, NumPy, SciPy
- Constructed POD-Galerkin reduced basis $\mathbf{X} \approx \Phi_r \mathbf{A}_r$ capturing 99.2% turbulent kinetic energy across 12 empirical modes from large-scale fluid mesh datasets.
 - Applied Sequential Thresholded Least Squares (STLSQ) extracting sparse nonlinear ODE governing dynamics $\dot{\mathbf{x}} = \Theta(\mathbf{x})\Xi$ with $1200\times$ computational speedup over full-order Navier-Stokes solvers.
 - Benchmarked multi-variate telemetry streams across 15,000+ spatial mesh points with zero numerical divergence across 15,000 time steps.

Autonomous Robotics, Embedded Firmware & Hardware Systems

6. Small Scale Raspberry Pi Based Autonomous Car ([IEEE ICRM 2025](#)) [First Author](#) · 2024 – 2025
C++17, OpenCV, Probabilistic Hough Transforms, PWM Closed-Loop PID, Asynchronous UDP, ARM Cortex-A
- Engineered real-time perception pipeline tracking lane boundaries at 35+ FPS on resource-constrained ARM Cortex-A SBCs.
 - Implemented asynchronous UDP telemetry server streaming video & telemetry frames with <15 ms round-trip latency.
 - Coupled optical incremental encoders with S-curve acceleration profiling, eliminating dynamic wheel slip across sudden turns.
7. Bluetooth Autonomous Mobile Robot with Soft-Start Velocity Profiling [GitHub/Bluetooth-Mobile-Robot](#) · 2024
Electrical Engineering Lab (23EEE105) · Arduino Uno R3, C++, Non-Blocking Asynchronous Loop, L298N Dual H-Bridge, HC-05 UART
- Developed non-blocking asynchronous time-based kernel (millis()) executing concurrent PWM motor control and Bluetooth command parsing.
 - Engineered handleSpeedTransition() algorithm ramping PWM duty cycles linearly (+5 PWM every 20 ms) for soft-start/soft-stop velocity profiling.
 - Eliminated mechanical gearbox backlash and high inductive current spikes on L298N dual H-bridge motor drivers under rapid direction reversals.
8. Industrial GSM & 4G LTE Remote Automation System [GitHub/Industrial-GSM-Automation](#) · 2024
Embedded C++, Arduino, A7670C 4G LTE/GSM Cellular Module, Non-Blocking AT Parser, 4-Channel 240V Relays
- Architected non-blocking AT command parser interfacing A7670C 4G LTE cellular module with microcontroller for worldwide SMS/cellular actuation.
 - Implemented sender phone number authentication/whitelisting and command decoding (s1on, s1off, status) with bidirectional SMS delivery feedback.

- Hardened 4-channel 240V opto-isolated relay module with RC snubber circuits for safe switching of heavy inductive appliances.

9. Secure Digital Ballot: Tamper-Proof Electronic Voting System

[GitHub/Secure-Digital-Ballot](#) · 2024

Pure C, Deterministic Finite State Machine, Custom Binary File Serialization (.bin), Struct Management

- Built deterministic tamper-proof voting engine in pure C utilizing custom binary file I/O serialization (.bin) to eliminate SQL injection vectors.
- Architected strict memory isolation between voter authentication roll and anonymous serialized ballot storage structs.
- Integrated real-time voter eligibility validation (<18 check), anti-double-voting logic, and password-authenticated administrative audit panel.

10. GATE IN & RA Master Command Center Preparation Studio

[GitHub/gate-in-ra-command-center](#) · 2024

Client-Side Web Platform · Pure HTML5, CSS3, Vanilla JavaScript, Page Visibility API, KaTeX Rendering

- Developed distraction-free GATE Robotics (RA) & Instrumentation (IN) study studio featuring Page Visibility API active foreground timer.
- Built interactive PYQ solver supporting numerical answer tolerance ($\pm 5\%$), step-by-step KaTeX mathematical proofs, and mistake diary.
- Integrated on-screen TCS iON scientific calculator simulator and 20-week structured syllabus diagnostic telemetry.

Experimental Thermodynamics & Field Energy Systems

11. 25-Day Coastal Solar Parabolic Dish Desalination Telemetry (Elsevier)

First Author / Co-Investigator · 2024

Precision RTD Sensors, Solar Pyranometer, Empirical Heat Transfer Modeling, Statistical Regression

- Executed 25-day field campaign logging solar irradiance, focal receiver temperature (116°C), and distillate yields.
- Achieved 99.5% Total Dissolved Solids (TDS) reduction (3,850 to 18 mg/L) and modeled empirical correlation (Adjusted $R^2 = 0.83$, $p < 0.05$).
- Authored complete manuscript submitted to Elsevier Desalination journal (Under Review).

KEYNOTE PRESENTATIONS, WORKSHOPS & SCIENTIFIC OUTREACH

- Technical Speaker, IEEE International Conference on Robotics and Mechatronics (ICRM 2025): Presented accepted research on vision-guided autonomous vehicles and real-time edge embedded control loops.
- Workshop Lead, Hands-On ROS 2 & Bare-Metal STM32 Systems Boot Camp (2024): Conducted 3-day technical boot camp training 40+ engineering undergraduates in CycloneDDS middleware, FreeRTOS task scheduling, and KiCAD schematic capture.
- Science Outreach Fellow, ISRO Space Exploration Day (2021): Delivered invited lecture on satellite orbital mechanics and atmospheric telemetry instrumentation to regional high school science scholars.

PROFESSIONAL REFERENCES & LABORATORY SUPERVISORS

- Available upon request (Faculty Mentors, Department of Robotics and Artificial Intelligence, Amrita Vishwa Vidyapeetham, Amritapuri Campus).